

Tutorial—2 Elasticity

Review:

- **Young's modulus, $Y = \text{linear stress} / \text{linear strain}$**

$$Y = (F/A) / (\Delta L/L)$$

where F is the normal force, A is cross sectional area, ΔL is change in length and L is the original length.

- **Rigidity modulus, $n = \text{tangential stress} / \text{tangential strain}$**

$$n = (F/A) / \theta$$

where F is the tangential force and θ is the tangential or shear strain.

- **Bulk Modulus, $\kappa = \text{Volume stress} / \text{volume strain}$**

$$\kappa = (F/A) / (\Delta V/V)$$

where ΔV is change in volume and V is original volume

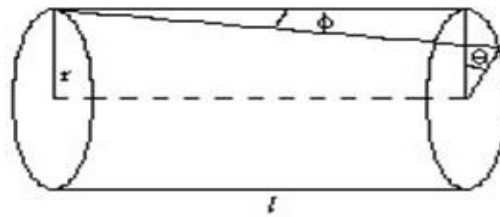
- **Poisson's Ratio, $\sigma = \beta/\alpha = \text{lateral strain} / \text{linear strain}$**

- **Couple per unit twist, $C = \pi n r^4 \theta / 2l$**

where r is the radius of the wire, l is length of the wire, θ is the angle of twist and n is the rigidity modulus.

- **Time period for torsional oscillation:**

$$T = 2\pi\sqrt{I/C} \quad \text{or} \quad C/4\pi^2 = I/T^2$$



- **Bending moment of a beam $= Y I_g / R$**

where R is radius of curvature of the bent beam and I_g is geometrical moment of inertia

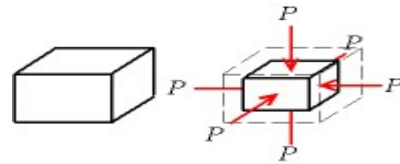
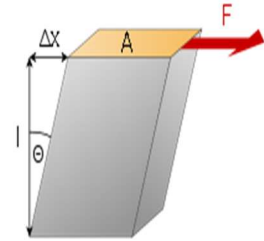
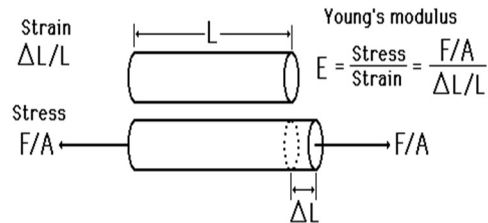
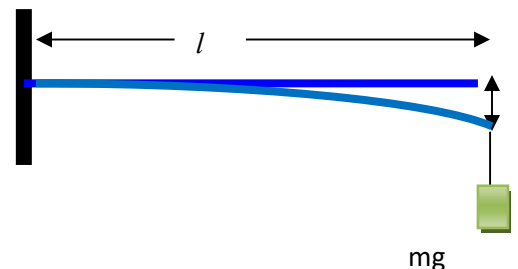
- $I_g = bd^3/12$ for rectangular cross section (b is breadth and d is thickness).
- $I_g = \pi r^4/4$ for circular cross section (r is radius)

- **Single Cantilever**

Young's modulus of a single cantilever,

$$Y = 4mg l^3 / bd^3 \delta$$

where mg is the load, l , b , d are length, breadth and thickness of the beam and δ is the deflection.



Numerical Problems

1. A steel wire of length 2.75 m and diameter 1 mm is fixed at the top and a load of 1 kg is suspended from the lower end. Calculate the elongation. Given the Young's modulus is $20 \times 10^{10} \text{ N/m}^2$.
2. A load of 3 kg produces an extension of 1.5 mm in a wire of length 4 m and diameter 1 mm. Calculate the Young's modulus of the wire.
3. What force is required to stretch a steel wire to double its length given the area of cross-section is 1.2 cm^2 and Y for steel is $20 \times 10^{10} \text{ N/m}^2$.
4. A copper wire has length 3 m and diameter 1 mm. Young's modulus and Poisson's ratio for copper are $12.5 \times 10^{10} \text{ N/m}^2$ and 0.26 respectively. The wire is fixed at one end and loaded with a weight of 10 kg at the other end. Calculate the elongation and lateral compression.
5. A solid copper cube has an edge of length 74.5 cm. How much stress must be applied to the cube to reduce the edge length 74 cm? (Given that $K_{\text{Cu}} = 1.4 \times 10^{10} \text{ N/m}^2$).
6. A material has Poisson's ratio 0.2. If a uniform rod of it suffers a longitudinal strain 4×10^{-3} , Calculate the percentage change in its volume.
7. What couple must be applied to a wire of length 1 m and diameter 1mm in order to twist one end through 90° when the other end is fixed? The rigidity modulus is $2.8 \times 10^{10} \text{ N/m}^2$.
8. An iron wire of length 1 m and radius 0.5 mm elongates by 0.32 mm when stretched by a force of 49 N, and twists through 0.4 radian when equal and opposite torques of $3 \times 10^{-3} \text{ N-m}$ are applied to its ends. Calculate the elastic constants Y and n of iron.
9. The free end of a wire of length 30 cm and diameter 0.56 mm passes through the center of a rectangular plate of length 20 cm and breadth 10 cm. The rigidity modulus of the material of the wire is $7.5 \times 10^{10} \text{ N/m}^2$. If the time period for torsional oscillations is 10 sec, calculate the couple per unit twist.
10. A cantilever undergoes a deflection of 10 mm under a certain load. What would be the deflection for the same load if the length and breadth are doubled and thickness is tripled?
11. A cantilever of length 0.5 m has a depression of 15 mm at its free end. Calculate the depression at a distance of 0.3 m from the fixed end.
12. A uniform rigid rod of length 1.2 m is clamped horizontally at one end. A weight of 1 kg is attached to the free end. Calculate the depression of a point 0.9 m distance apart from the clamped end. The diameter of the rod is 2 cm. Young's modulus of the material of the rod is $1.013 \times 10^{10} \text{ N/m}^2$.
13. A cantilever with 3cm breadth, 1.2m length and 10mm width is fixed at one end. A load of 1.7kg is applied at the other end. Calculate the depression at a distance 0.9m from the fixed end and free end if the Young's modulus is $11 \times 10^{10} \text{ N/m}^2$.
14. Calculate the weight at the end of a single cantilever of length 40 cm, breadth 3 cm and thickness 2 mm such that the arc at the middle of the beam has a radius of curvature of 0.5 m. (given $Y = 20 \times 10^{10} \text{ N/m}^2$).
15. A uniform rigid rod of length 1 m is clamped horizontally at one end. A weight of 1.5 kg is attached to the free end. Calculate the depression of the midpoint of the rod. The diameter of the rod is 1.75 cm and the young's modulus of the material of the rod is $1 \times 10^{10} \text{ Nm}^{-2}$.
16. What couple must be applied to a wire of length 1 m and diameter 0.5mm in order to twist one end through 20° when the other end is fixed. The rigidity modulus is $3 \times 10^{10} \text{ N/m}^2$. Also calculate the shearing angle.
17. A wire with 3m length for which Young's modulus is $10 \times 10^{10} \text{ N/m}^2$ has a 1mm diameter. A load of 10kg is attached to one end, how much extension is observed. If Poisson's ratio is 0.26 what is the lateral compression produced.
18. Calculate the weight at the end of a single cantilever of length 30 cm, breadth 2.5 cm and thickness 1 mm such that the radius of curvature at the distance 10cm from free end is 1 m. (Given $Y = 18 \times 10^{10} \text{ Nm}^{-2}$)
19. A cantilever of length 1m has a depression of 10mm at its free end. Calculate the depression at midpoint of the cantilever.